MODULE - 3 :- TRIGONOMETRY.

Circular and Hyperbolic Functions of a Complex Variable

Circular Functions.

When x is real,

$$e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots$$

NOTES

$$e^{x+y} = e^x \cdot e^y$$

$$\cos n\pi = (-1)^n, \quad n \in \mathbb{Z}$$

$$\sin n\pi = 0, \quad n \in \mathbb{Z}$$

Result

$$e^{i\theta} = \left(1 + \frac{i\theta}{1!} + \frac{(i\theta)^2}{2!} + \frac{(i\theta)^3}{3!} + \dots\right)$$

$$= 1 + i\theta - \frac{\theta^2}{2!} - \frac{i\theta^3}{3!} + \frac{\theta^4}{4!} - \dots$$

$$= \left[1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \frac{\theta^6}{6!} + \dots\right] + i\left[\theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \dots\right]$$

$$e^{i\theta} = \cos\theta + i\sin\theta \quad \text{—— (1)}$$

$$e^{-i\theta} = \cos\theta - i\sin\theta \quad \text{—— (2)}$$

① + ② $\Rightarrow e^{i\theta} + e^{-i\theta} = 2\cos\theta$

$\Rightarrow \boxed{\cos\theta = \frac{e^{i\theta} + e^{-i\theta}}{2}}$

① − ② $\Rightarrow e^{i\theta} - e^{-i\theta} = 2\sin\theta . i$

$\Rightarrow \boxed{\sin\theta = \frac{e^{i\theta} - e^{-i\theta}}{i}}$

Result

For all values of x, real or complex, we've

$\cos x = \frac{e^{ix} + e^{-ix}}{2}$, $\sin x = \frac{e^{ix} - e^{-ix}}{2i}$

Thes results are known as Euler's exponential values.

Q. Let x, y be real or complex then prove that

i) $\sin(x+y) = \sin x \cos y + \cos x \sin y$

ii) $\cos(x+y) = \cos x \cos y - \sin x \sin y$

iii) $\sin(x-y) = \sin x \cos y - \cos x \sin y$

iv) $\cos(x-y) = \cos x \cos y + \sin x \sin y$

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i) We know that $\cos x = \frac{e^{ix} + e^{-ix}}{2}$, $\sin x = \frac{e^{ix} - e^{-ix}}{2i}$

RHS = $\sin x \cos y + \cos x . \sin y$

$= \frac{e^{ix} - e^{-ix}}{2i} \times \left(\frac{e^{iy} + e^{-iy}}{2}\right) + \left(\frac{e^{ix} + e^{-ix}}{2}\right) \times \left(\frac{e^{iy} - e^{-iy}}{2i}\right)$

$$= \frac{e^{i(x+y)} + \cancel{e^{i(x-y)}} - \cancel{e^{-i(x-y)}} - e^{-i(x+y)} + e^{i(x+y)} - \cancel{e^{i(x-y)}} + \cancel{e^{-i(x-y)}} - e^{-i(x+y)}}{4i}$$

$$= \frac{2e^{i(x+y)} - 2e^{-i(x+y)}}{4i}$$

$$= \frac{e^{i(x+y)} - e^{-i(x+y)}}{2i}$$

$= \sin(x+y)$

$= \underline{\underline{LHS}}$

(ii) RHS $= \cos x \cos y - \sin x \sin y$

$$= \left(\frac{e^{ix} + e^{-ix}}{2}\right) \cdot \left(\frac{e^{iy} + e^{-iy}}{2}\right) - \left(\frac{e^{ix} - e^{-ix}}{2i}\right) \times \left(\frac{e^{iy} - e^{-iy}}{2i}\right)$$

$$= \frac{2e^{i(x+y)} + 2e^{-i(x+y)}}{4} = \frac{e^{i(x+y)} + e^{-i(x+y)}}{2}$$

$= \underline{\underline{\cos(x+y)}}$

Illy (iii), (iv).

Periods of Complex Circular Functions

$$\sin(x+2\pi) = \sin x \cdot \cos 2\pi + \cos x \sin 2\pi$$

$$= \sin x \cdot 1 + \cos x \times 0$$

$$= \sin x$$

|||ly $\cos(x+2\pi) = \cos x$

Hence $\sin x$ and $\cos x$, both remain the same when x is increased by 2π.

|||ly the will remain the same when x is increased by $4\pi, 6\pi, \ldots, 2n\pi$.

→ A periodic function of period p is

$$f(x \pm p) = f(x)$$

Imp Q. P.T for all values of x and y, real or complex

i) $\cos^2 x + \sin^2 x = 1$

$$LHS = \cos^2 x + \sin^2 x$$

$$= \left(\frac{e^{ix} + e^{-ix}}{2}\right)^2 + \left(\frac{e^{ix} - e^{-ix}}{2i}\right)^2$$

$$= \frac{e^{2(ix)} + 2e^{ix} \cdot e^{-ix} + e^{-2(ix)} - (e^{2(ix)} - 2e^{ix} \cdot e^{-ix} + e^{-2(ix)})}{4}$$

$$= \frac{\cancel{e^{-x}}+2+\cancel{e^{x}}-(\cancel{e^{-x}}-2+\cancel{e^{x}})}{4}$$

$$= \frac{\cancel{e^{-x}}+2+\cancel{e^{x}}-\cancel{e^{-x}}+2-\cancel{e^{x}}}{4}$$

$$= 4/4 = \underline{\underline{1}} = RHS.$$

$$= \frac{(e^{ix})^2+2e^{ix}\cdot e^{-ix}+(e^{-ix})^2-(e^{ix})^2+2e^{ix}\cdot e^{-ix}-(e^{-ix})^2}{4}$$

$$= 4/4 = 1 = \underline{\underline{RHS}}$$

ii) $\underline{\cos(-x) = \cos x}$

$$LHS = \cos(-x) = \frac{e^{-ix}+e^{ix}}{2}$$

$$= \frac{e^{ix}+e^{-ix}}{2} = \underline{\underline{\cos x}} = RHS$$

iii) $\underline{\sin(-x) = -\sin x}$

$$LHS = \sin(-x)$$

$$= \frac{e^{-ix}-e^{ix}}{2i}$$

$$= -\frac{(e^{ix}-e^{-ix})}{2i}$$

$$= -\sin x = \underline{\underline{RHS}}$$

iv) $\underline{\cos 2x = \cos^2 x - \sin^2 x}$

$$RHS = \cos^2 x - \sin^2 x$$

$$= \left(\frac{e^{ix}+e^{-ix}}{2}\right)^2 - \left(\frac{e^{ix}-e^{-ix}}{2i}\right)^2$$

$$= \frac{e^{2ix}+2e^{ix}\cdot e^{-ix}+e^{-2ix}}{4} + \frac{\left(e^{2ix}-2e^{ix}\cdot e^{-ix}+e^{-2ix}\right)}{4}$$

$$= \frac{2e^{2ix}+2e^{-2ix}}{4} = \frac{2e^{2ix}+2e^{-2ix}}{4}$$

$$= \frac{e^{2ix}+e^{-2ix}}{2}$$

$$= \cos 2x$$

v) $\cos 2x = 1 - 2\sin^2 x$

$$RHS = 1 - 2\sin^2 x$$

$$= 1 - 2\left(\frac{e^{ix}-e^{-ix}}{2i}\right)^2$$

$$= 1 + \frac{2}{4}\left(e^{2ix} - 2\cdot e^{ix}\cdot e^{-ix} + e^{-2ix}\right)$$

$$= 1 + \frac{1}{2}\left(e^{2ix} - 2 + e^{-2ix}\right)$$

$$= \frac{\cancel{2}+e^{2ix}-\cancel{2}+e^{-2ix}}{2} = \frac{e^{2ix}+e^{-2ix}}{2} = \cos 2x$$

vi) $\sin 3x = 3\sin x - 4\sin^3 x$

$$RHS = 3\sin x - 4\sin^3 x$$

$$= 3\left(\frac{e^{ix} - e^{-ix}}{2i}\right) - 4\left(\frac{e^{ix} - e^{-ix}}{2i}\right)^3$$

$$= \frac{3e^{ix} - 3e^{-ix}}{2i} + 4\left(\frac{e^{3ix} - 3e^{i2x}\cdot e^{-ix} + 3e^{ix}\cdot e^{-i2x} - e^{-i3x}}{28i}\right)$$

$$= \frac{3e^{ix} - 3e^{-ix} + e^{3ix} - 3e^{ix} + 3e^{-ix} - e^{-i3x}}{2i}$$

$$= \frac{e^{3ix} - e^{-i3x}}{2i} = \sin 3x = RHS$$

Q. $\cos x - \cos y = 2\sin\left(\frac{x+y}{2}\right)\cdot\sin\left(\frac{y-x}{2}\right)$

$$RHS = 2\sin\left(\frac{x+y}{2}\right)\cdot\sin\left(\frac{y-x}{2}\right)$$

$$= 2\left[\frac{e^{i\left(\frac{x+y}{2}\right)} - e^{-i\left(\frac{x+y}{2}\right)}}{2i}\right] \times \left[\frac{e^{i\left(\frac{y-x}{2}\right)} - e^{-i\left(\frac{y-x}{2}\right)}}{2i}\right]$$

$$= -\frac{1}{2}\left[e^{i\left(\frac{x+y}{2}\right)}\cdot e^{i\left(\frac{y-x}{2}\right)} - e^{i\left(\frac{x+y}{2}\right)}\cdot e^{-i\left(\frac{y-x}{2}\right)} - e^{i\left(\frac{x+y}{2}\right)}\cdot e^{i\left(\frac{y-x}{2}\right)} + e^{-i\left(\frac{x+y}{2}\right)}\cdot e^{-i\left(\frac{y-x}{2}\right)}\right]$$

$=$

$$= -\frac{1}{2}\left(e^{iy} - e^{ix} - e^{-ix} + e^{-iy}\right)$$

$$= \frac{e^{ix} + e^{-ix}}{2} + \frac{e^{iy} - e^{-iy}}{-2}$$

$$= \cos x - \cos y = LHS$$

HW

Q. $\sin x - \sin y = 2\cos\left(\frac{x+y}{2}\right) \cdot \sin\left(\frac{x-y}{2}\right)$

× NOTE

If x is a purely imaginary quantity, i.e., iy :

then, $\sin x = \sin(iy)$

$$= \frac{e^{i(iy)} - e^{-i(iy)}}{2i} = \frac{-(e^{y} - e^{-y})}{2i}$$

$$= i\,\frac{(e^{y} - e^{-y})}{2}$$

$\cos x = \cos(iy)$

$$= \frac{e^{i(iy)} + e^{-i(iy)}}{2} = \frac{e^{-y} + e^{y}}{2}$$